Response to Reviewers

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Point-by-point response to reviewers

Reviewer 1

This is a good paper that addresses some important issues with mobile phone data (MPD) in assessing the effects of shocks. I have a number of concerns that should be addressed before it is ready for publication, however.

Motivation and base case

Reviewer 1, Comment 1 (R1.1). *First, at a meta level, I wonder about impact. Many researchers working with MPD will have already shown population-coverage correlations and looked at demographic biases in deviations/residuals. Because many reviewers/researchers will address bias in the SI of their papers, I think the study would benefit from more work on whether or not the typical strategies are insufficient. This could be as simple as using a linear model, residualising coverage on population and looking at demographic/spatial biases in the residuals. I think the strength of this study actually lies in the spatial dimension, since most data validations will not include checks for spatial autocorrelation on the residuals of this coverage population model, but these are important as diagnostics that indicate unmodelled spatial structure, whether or not there might be some sort of unobserved/unmodelled bias in the data. Building this "base case" and finding out if it is insufficient, then, represents an important first step in improving the paper.*

> **A:** We agree that some studies working with digital trace data have assessed data validity exploring linear correlation. Additionally, if the sample composition is known (e.g. age/sex/group shares), adjustment methods such as post-stratification can be applied. However, MPD records are usually anonymised and do not include the demographic composition required for those corrections. In these cases, validation is typically limited to aggregate checks based on the relationship between digital-trace and benchmark population counts, through simple linear correlation measures (Iradukunda, Rowe, and Pietrostefani 2025; Renninger and Cabrera 2025; Wang et al. 2022; Moro et al. 2021; Cabrera et al.

2025). This provides a useful "sanity check" because it confirms proportional scaling, i.e. more populated areas tend to generate more digital traces, but it implicitly assumes that $P_i^D = \alpha P_i$, where P_i^D is the digital-trace population count in area i and P_i is the benchmark population from e.g. the census. Under this assumption, high correlation or high R^2 is often interpreted as evidence of representativeness. Given our definition of coverage bias $e_i = 1 - P_i^D / P_i$, the assumption also implies constant bias $e_i^0 = 1 - \alpha$.

Following the reviewer's recommendation, we implement this base case explicitly at Local Authority Distric (LAD) level by estimating $P_i^D = \hat{\alpha} P_i$. We then compute residual bias as $r_i = e_i - (1 - \hat{\alpha})$, where e_i is the observed coverage bias. We use these residuals to assess what the conventional proportional baseline leaves unexplained, including residual spatial dependence (Moran's I of residuals) and dependence on contextual covariates (linear and nonlinear associations with area-level covariates). This allows us to better justify the need for the proposed framework based on a more flexible, machine learning approach.

To keep the narrative logic of the paper, we add this analysis as a "baseline case" rather than part of the proposed methodological framework. We use it to show "what is typically done", why the conventional approach is not sufficient on its own and why our framework is needed. We present the main base-case findings in a subsection in the "Results" section before the framework results, and place the full technical details and baseline case outputs in the Supplementary Information.

Action: Particularly, we have taken the following actions in response:

- We updated the text in the "Introduction" explaining that conventional correlation/proportionality checks are a useful baseline but may miss fundamental differences in the sources of bias. Particularly, we updated the following paragraph: *Efforts have been made to assess bias in aggregate population counts from digital data sources.When the demographic composition of digital data samples is available, post-stratificationtechniques can be applied to identify and adjust group-level imbalances [30,31]. Based on user'sdemographic information, previous studies have been able to establish systematic gender, ageand socioeconomic biases in data obtained via API (or Application Programming Interface)from social media platforms, such as Facebook and Twitter/X, often at the national scale.However, MPD-based records are generally anonymised and do not include the required samplecomposition attributes for post-stratification. In such cases, data validation is typically limitedto aggregate proportionality assessment between digital and census counts across geographicalareas [17,22,32,33]. While this approach is useful as a first diagnostic tool, strong globalgeographic proportionality does not ensure that the sample is representative across populationgroups or places [34,35]. What is therefore needed is a systematic approach that can evaluaterepresentativeness when sample composition is unknown and explicitly quantify subnationalheterogeneity in bias.*
- We added an explicit linear baseline model ($P_i^D = \hat{\alpha} P_i$) and residual bias definition in the Methodology, to illustrate the conventional approach to assessing representativeness of digital trace data and explain why this is insufficient.

- We revised the "Results" subsection entitled "Population coverage bias varies widely over space" to report baseline case fit (R^2). The results are reflected in the scatter plots of Figure 3.
- We extened the "Results" section adding a new subsection for an assessment of variability in residuals resulting from the baseline case. Particularly, we mapped the residuals, computed the Moran's I of residuals and assessed the residual dependence on contextual covariates, including linear and nonlinear specifications. These results are summarised in Figure 4.
- We now provide a complete assessment of residual variability in the Supplementary Tables 3-7. Table 3 is a baseline summary table where, for each data source, we report the proportional-model estimate ($\hat{\alpha}$), model fit (R^2), and Moran's I (with p-value) for baseline residuals. Tables 4-7 show full pairwise residual–covariate results across all candidate contextual variables. For each covariate, we report Pearson, Spearman, distance correlations with residual bias, and compare linear (OLS) versus nonlinear (GAM, $k = 5$) specifications using AIC and BIC. Across data sources, GAM specifications generally result in lower AIC and BIC than linear models, indicating that residual bias is more consistently explained by nonlinear contextual relationships. These supplementary results support our conclusion that proportional validation is insufficient to assess representativeness of digital trace data.

Bias measure

R1.2: *The authors present a measure of bias e_i that represents the deviations from perfect coverage They suggest that a situation wherein there more mobile devices that people within a given areal unit, such that $1 - c_i < 0$, is rare, so the measure functionally bounded [0, 1]. Because it could happen, though, a mention of how it is handled is also important—eg winsorisation/censoring.*

A: The reviewer is correct that our definition of coverage $c_i = P_i^D/P_i \times 100$ allows $c_i > 100$ when the observed number of devices or users exceeds the census population in an area, which would yield negative values of our bias measure $e_i = 100 - c_i$. While this situation is theoretically possible (e.g. due to device duplication, temporal mismatch or other measurement artefacts), it does not occur in our empirical application. We therefore do not apply winsorisation or censoring. In principle, allowing e_i to take negative values is methodologically acceptable and interpretable as "over-coverage" relative to the census benchmark. Our modelling approach can accommodate such values.

Action: We revised the manuscript to explicitly clarify the domain and handling of the coverage-bias metric. Particularly, we have updated the Methods subsection "Measuring population coverage bias". We added the following text: *"In principle, c_i is not strictly bounded above by 100. Values above 100 can occur..."* and *"Positive values ($e_i > 0$) indicate under-coverage... while negative values ($e_i < 0$) indicate over-coverage... Negative values... are therefore valid and can be handled in our empirical approach..."*

R1.3: *Another problem is that the bias might be subtler than that because coverage is often low that even devices doubled or tripled in some households would still not push* $c_i > 1$*: that is, because these samples are often on the order 10% of the population, there could still be many duplicates per household. I do not think there is a good solution here, and likely only exists for MPD, not for Facebook/Twitter, but I think the authors should comment on it and include a check for devices that display X% similarity. A complete process for finding duplicates is beyond the scope of the paper but it could be an important source of bias so I also do not think we can write it off out of hand; some parsimonious check would help.*

A: We agree - multi-device ownership (or duplication) can drive the prevalence of coverage bias. Duplication may artificially inflate device-based population counts and lower coverage bias. Individuals may own multiple devices or generate multiple identifiers, and if this may vary systematically across areas contributing to spatially uneven overestimation. Our proposed methodology do not seek to identify individual sources of data biases, including multi-device ownership. As stated in the manuscript, our framework focuses on quantifying the extent of "cumulative'' bias; that is, resulting bias from the accumulation of these error sources, including multi-device ownership. We do this because our ultimate aim is to develop a generalisable methodology that can be applied to spatially aggregated data from any source or context, recognising that this is the main format that data providers are used to share data in compliance with ethical and privacy regulations. In our analysis, over-representation of wealthy households in the data is thought to partly reflect higher incidence of smartphone penetration among these families. Yet, it is hard to find evidence on the prevalence of multi-smartphone ownership.

Action: We revised the manuscript to indicate: (1) that device duplication is a potential problem; (2) that our approach does not seek to isolate and measure it; and, (3) that our framework focuses on quantifying the prevalence of "cumulative" bias. Specifically, we:

- Added text in "Methods" subsection "Measuring population coverage bias" to indicate that multi-device ownership or duplication can occur, even when coverage is below 100%. Our approach does not seek to identify individual sources of biases, such as multi-device ownership. It focuses on quantifying the prevalence of "cumulative" bias, including multi-device ownership.
- Revised the text at the beginning of the "Data" section to indicate that duplication analysis requires access to individual-level records. However, mobile phone data at such level are rarely available due to privacy and confidentiality concerns. In our case, data access across sources is asymmetric. Only Multi-app1 is available in raw format, while Meta, Twitter/X and Multi-app2 are available as aggregated, privacy-preserving inputs. We indicate that a consistent record-level deduplication check cannot be implemented across all datasets.

Model

R1.4: *Given that the paper explores potential spatial biases in the data, the k-fold cross-validation should include spatial folds as well; k-fold with contiguity clusters would help and something with regions (like leaving London out) would be even stronger.*

A: We appreciate this suggestion and agree that, when spatial dependence is present, a model that incorporates spatial cross-validation might be more appropriate. Following the recommendation, we implemented spatial block cross-validation and compared the resulting models against those estimated under the original random holdout approach.

To assess the practical relevance of implementing spatial cross-validation, we examined Moran's I for both the dependent variable and residuals of our fitted models. These analyses show that spatial autocorrelation in the dependent variable is generally very weak across datasets, and residual spatial autocorrelation is small and statistically non-significant. This indicates that strong spatial leakage is not a dominant feature of our data. We clarify that this may be due to the use of Local Authority Districts and that spatial autocorrelation may emerge as an issue if smaller geographical units are used for analysis.

We also assessed spatial autocorrelation in the contextual covariates and found that some covariates display spatial dependence . We therefore proceeded with spatial block cross-validation. However, when comparing results from random and spatial block cross-validation, we do not find clear evidence that one validation approach systematically outperforms the other. Out-of-sample performance metrics and residual Moran's I are comparable and the conclusions of the paper, regarding the main contextual drivers of coverage bias, remain stable across validation approaches.

We therefore present spatial block cross-validation as a robustness test rather than as a superior validation strategy in the present study. We retain the simpler random holdout specification in the main text for interpretability and report the spatial block cross-validation results in the Supplementary Information. This choice is also consistent with recent debate in the literature, where spatial cross-validation has been recommended as a more conservative choice when spatial dependence may be present, but its universal use has also been questioned. Yet, we again emphasise that spatial autocorrelation is likely to emerge at lower geographical scales as the spatial interactions between places through mobility flows is more fluid.

Action: We have taken the following actions in response:

- We assessed spatial autocorrelation of model covariates and report the outcomes in Supplementary Table 2.
- We implemented spatial block cross-validation models for all datasets. We report this implementation as an additional robustness check alongside the original random cross-validation. The model performance is now reported in the Supplementary Table 8 and compared to random cross-validation, considering the out-of-sample root mean squared

error, Pearson correlation, Spearman correlation and standard deviation of prediction error.
- We computed and reported Moran's I of model residuals to assess whether meaningful residual spatial dependence remained after model fitting. We did so for all model specifications and included these values in Supplementary Table 8.
- We added Supplementary Figure 2, which shows that the patterns of variable importance are consistent across validation strategies.
- We revised the manuscript text in the Methods (subsection "Identifying sources of representativeness bias via explainable machine learning") and Discussion (in the fifth paragraph) to clarify that spatial block cross-validation is used here as a robustness check motivated by the possibility of spatial dependence.
- We also added text in Section 3(c) to state that while we found no statitically significant evidence of spatial dependence, it is likely to occur a finer levels of spatial aggregation.

R1.5: *The authors also opt out of using spatial lags in the model, because there is not strong autocorrelation in the DV, but this is probably the wrong call: a lack of autocorrelation in the DV does not guarantee that the residuals are not autocorrelated. Further, because this approach is automated, using regularisation to determine what makes the cut, it makes more sense to include lags (of predictors) and let ML do the work of figuring out if they are important. Please also report Moran's I of residuals.*

A: We agree that weak spatial autocorrelation in the dependent variable does not guarantee the absence of spatial dependence in model residuals. In response, we computed Moran's I for model residuals across model specifications. These tests show that residual spatial autocorrelation is generally weak and, in most cases, not statistically significant. We also assessed spatial autocorrelation in the model covariates and found that some covariates do display substantial spatial structure. This latter finding provides motivation for testing whether model results are sensitive to the inclusion of spatially lagged covariates.

Following the reviewer's suggestion, we therefore estimated additional model specifications including spatially lagged covariates, under both random and spatial block cross-validation. We then compared these specifications with the original non-lagged models in terms of out-of-sample performance and patterns of variable importance. Across datasets, the inclusion of lagged covariates does not necessarily improve model performance. Additionally, the inclusion of lagged covariates does not alter the main conclusions regarding the leading contextual drivers of population coverage bias.

For this reason, we treat models with spatially lagged covariates as robustness tests. We report these results in the Supplementary Information and clarify in the manuscript that the main interpretation is robust to whether lagged covariates are included.

Action: We have taken the following actions in response:
- We assessed spatial autocorrelation in the contextual covariates and report the corresponding Moran's I statistics in Supplementary Table 2.

- We estimated additional model specifications including spatially lagged covariates under both random and spatial block cross-validation. We compared lagged and non-lagged specifications using out-of-sample performance metrics and variable-importance profiles. These are reported in Supplementary Table 8. We also computed and report Moran's I of model residuals for the machine-learning models in order to assess whether spatial dependence remains after model fitting. These results are also provided in Supplementary Table 8.
- We revised the manuscript text to clarify that models including spatially lagged covariates are used as robustness checks, and that the substantive conclusions are stable across spatially lagged and non-lagged specifications.

Less important

R1.6: *There is no mention of how many LADs are in England but it seems small. XGBoost is fit for purpose here, but including hyperparameters to alleviate any concerns about overfitting would be good. Please report capacity controls; add a brief calibration plot (predicted vs. observed) and residual maps.*

A: We appreciate this suggestion and agree that additional reporting of model configuration and performance improves transparency and reproducibility. In response, we now report the number of local authority districts included in the analysis (331), provide the selected hyperparameters for the XGBoost models in the Supplementary Information and report a larger set of model performance metrics, including RMSE, Pearson and Spearman correlation of predicted vs. observed data, the standard deviation of prediction errors and Moran's I of model residuals. We used a grid of hyperparameters to optimise and select our final set of hyperparameters. This is all documented in the code that we will make available in a Github repository. Additionally, we included maps of the XGBoost model residuals in the Supplementary Information Figure 3.

Action: We have revised the manuscript in the following ways:

- We now report the number of local authority districts included in the analysis: 331. The number can be found in the second paragraph of the "Data" section.
- We report the key XGBoost hyperparameters used to control model capacity in Supplementary Table 9. These include maximum tree depth, learning rate (eta), subsample ratio, column subsampling and L1 and L2 regularisation penalties. These parameters were tuned using random holdout and spatial block cross-validation and selected based on minimising out-of-sample RMSE. We also indicate that regularisation and early stopping criteria were used to prevent excessive tree growth and mitigate overfitting.
- We expanded model evaluation to report RMSE, Pearson correlation, Spearman correlation, the standard deviation of prediction error and Moran's I of model residuals. These metrics are reported in Supplementary Table 8.
- We included Figure 3 in the Supplementary Information displaying our model residuals.

R1.7: *Figure 4 looks great but "Lower supervisory" should be put either at noon or 6 (as it were, on the clock) so that it doesn't protrude so far off to the side. As it is, that text pushes all the other figures off to the left side.*

Action: We have now revised the Figure so the "Lower supervisory" and other label do not protrude awkwardly.

Reviewer 2

The paper presents a national-scale assessment of bias in social media and multi-source mobile phone application data by comparing these sources to the census, then linking patterns to area characteristics using an interpretable machine-learning approach. The study is ambitious, the analysis appears robust, and the figures and overall presentation are mostly strong. A revision that clarifies a few key points would strengthen the paper and make the contribution easier to interpret and reuse.

Reviewer 2, Comment 1 (R2.1): *At the moment, the paper reads more strongly as a set of findings for these specific data sources in the UK than as a general workflow. This is mainly because the home-area approach builds on existing methods and the analysis is then carried out at Local Authority District (LAD) level, which is higher than the scale many studies would use to link census/IMD variables or to adjust for bias. It may help to either frame the contribution more clearly as UK-wide evidence, or explain more directly how and why the approach would transfer to other places and finer geographic scales.*

A: Thanks for raising these points. Our paper has two contributions: methodological and substantive. We propose a general framework to assess population coverage bias in aggregated mobile phone application data when individual-level demographic attributes are unavailable. The UK-specific analysis is the substantive contribution, and it is intended to illustrate how our general framework can be used in a realistic applied setting.

This choice of working at the Local Authority District (LAD) level was constrained by data harmonisation and comparability across our four sources of digital trace data. LADs are the finest common geography consistently available across the four sources. The choice is therefore a feature of this empirical application and not a constraint of the proposed framework, which can in principle be implemented at finer spatial resolutions or in other national contexts wherever an external population benchmark and area-level covariates are available.

Action: To make this clearer we have taken the following actions:

- We revised the "Introduction" to more explicitly define our main methodological and substantive contributions. We added the following text in the last two paragraphs describing the contributions: *"Our methodological framework is not specific to the UK case study, but is designed to be applicable across different national contexts and spatial scales wherever a suitable benchmark population and linked area-level covariates are available.*

We applied our approach to UK data, to illustrate how our proposed approach can be applied to empirical data."

- We clarified in the "Data" section that the use of LADs reflects cross-source harmonisation and data availability constraints, rather than a requirement of the proposed framework. Particularly, we added the following text: *"To enable cross-source comparison, we harmonise all datasets to the finest common geography that is consistently available across the four sources used in our study. Our use of local authority areas reflects data availability and cross-source comparability constraints and not a requirement of the proposed framework. The analytical sample includes 331 LADs. The same workflow can be applied in other national contexts and at finer spatial resolutions, such as smaller administrative units or regular grids, provided that a comparable benchmark population and linked area-level covariates are available".*
- We expanded the "Discussion" to note that the framework is designed to be adaptable across geographies, including settings where alternative benchmark populations, such as gridded population datasets, may be used. In particular, we added the following text: *"However, the proposed framework is designed to be adaptable across geographic contexts provided that a suitable benchmark population and linked area-level covariates are available. In settings where recent census benchmarks are unavailable, future applications could draw on gridded population datasets such as WorldPop (Tatem, 2017).*

R2.2: *The choice to work at LAD level needs a clearer justification and a fuller discussion of what it means for interpretation. LADs are useful for a national overview, but many common uses of these data (including mobility analysis) rely on smaller areas. The paper notes that results may differ at other geographies (page 14, line 9); expanding this would help readers understand whether the LAD patterns are likely to hold when broken down into smaller areas. The home-area workflow also needs a little more detail: the paper should state the geography at which home area is first estimated (before aggregation to LAD), whether the same method is used for multi-source dataset 1 and dataset 2, and what "privacy preserving" means in practice for this study.*

A: As described above, we use LADs was to analyse our four data sources based on a common spatial framework to ensure consistency and comparability. We appreciate this rationale was missing from the original text. We have revised the text to explicitly state this, explaining that LADs represented the finest common geography that we could used to establish a consistent spatial framework for the analysis of the four data sources used in the study. We also revised the text to indicate that this is not a limitation of our proposed framework, but it is a constraint in our empirical application of the framework to the UK. We also revised the text to discuss that LAD-level patterns should be interpreted as overall national-scale comparative evidence and patterns may not map to finer spatial units. Unobserved patterns may become visible as local heterogeneity can be identifiable.

Additionally, we have expanded the description of the home-area identification. We now state more clearly the geography at which home locations are first inferred before aggregation to LAD. We describe the procedures used to infer home location and explain what we mean by "privacy-preserving" in the context of this study. Specifically, we revised the text to emphasise that our framework seek to leverage the use of anonymised and aggregated data products, and that it has been designed to avoid the need for identifiable individual-level records. We appreciate that this may be at the expense of precision. At the same time, we also recognise that individual-level records are rarely accessible. Instead, anonymised and aggregated data products have become the standard format used by mobile network operators and data brokers to facilitate access to mobile phone data.

Action: We have revised the manuscript in the following ways:

- We have clarified in the "Data" section that LADs were selected because they provide the finest common geography available across all four data sources. We do this to enable consistent comparisons across our four data sources.
- In the "Discussion", we expanded the argument around the interpretation of results given our reliance on LADs as unit of analysis. We explain that the LAD-level results should be interpreted as overall national-scale comparative evidence. Results may differ when analysed at finer spatial resolutions as currently unobserved patterns may become visible in the data. Particularly, we have added the following text: *"the results presented here at local authority districts should be interpreted as overall national-scale comparative evidence across harmonised spatial units. Currently unobserved patterns may become visible at finer geographic scales unveiling within-area heterogeneity, and the magnitude or configuration of coverage bias may differ."*
- We added a description on the definition of home area. This includes identification of the geography at which home is first inferred for Multi-app1, and the preprocessing and aggregation level available to us for Multi-app2. The added text can be found in the following sections: "Data - Multi-app1" and "Data - Multi-app2".
- We have added text to the opening paragraphs of the "Data" section explaining what we mean by privacy-preserving data. Particularly, we added: *"In this study, "privacy-preserving" refers to the use of data that has been anonymised, spatially and temporally aggregated, and that does not provide access to identifiable individual-level records.".*

R2.3: *Population coverage measures should be defined more clearly. In particular, it should be stated whether the denominator is the adult population only, given that GDPR mobile app products typically cover users aged 16+. If children are included in the denominator, that should be acknowledged as a limitation. It would also be useful to note how differences in the adult share across areas might affect equation 3.1 and later steps that use "population coverage bias".*

A: In our analysis, the denominator corresponds to the total resident population (all ages) as reported in the 2021 Census. We acknowledge that, in principle, if mobile application datasets disproportionately represent individuals aged 16+, including children in the denominator could increase the estimated coverage bias in areas with a larger share of young people.

However, we do not observe individual-level age information in the digital trace datasets and therefore cannot verify the age composition of the captured population. Redefining the denominator as the 16+ population would therefore impose an assumption that cannot be empirically validated with the available data. For transparency and comparability across sources, we retain the total resident population as the benchmark.

We further agree that spatial variation in age structure may influence Equation (3.1) and subsequent modelling stages. To account for this, age composition variables are included among the area-level predictors in the explainable machine learning framework, allowing the model to assess whether differences in adult population share systematically contribute to observed coverage bias.

Action: We have taken the following actions:

- We have clarified in the "Methods" section that the denominator in the population coverage calculation corresponds to the total resident population (all ages) from the 2021 Census. Particularly, we added the following text: *"We use the total resident population, rather than the adult population only, as a common benchmark across all datasets. Mobile application datasets may disproportionately represent individuals above a certain age (e.g. 16+). In that case, including children in the denominator may increase estimated coverage bias in areas with lower adult population shares. However, givent that age is not observed in the digital trace data, we cannot verify the age composition of the captured population and therefore cannot reliably restrict the denominator to the adult population. This however could be thought as a parameter in our framework and the denominator could be restricted to specific age ranges as analysts may find it most appropriate".*

R2.4: *In a few places, the text would benefit from clearer separation between coverage, bias, and representativeness. Higher penetration for Facebook and X does not necessarily mean these sources are representative, and multi-source app data may have lower coverage but still be more balanced across some groups (as suggested in prior UK evidence). This distinction is mentioned in the discussion, but it should be applied more consistently in the Results and Discussion, especially in section 4b where social media show higher coverage.*

A: As recognised in our original submission, we agree that higher coverage does not imply representativeness and this should be emphasised. A source may achieve relatively high penetration and still misrepresent particular demographic, socioeconomic or geographic groups. Conversely, a source with low overall coverage may provide a balanced representation across some population groups. In the revised manuscript, we have added text to make a clearer distinction between overall population coverage, population coverage bias and representativeness. We also added text to link this distinction more explicitly to the proportional baseline analysis introduced in response to Reviewer 1, Comment 1. Specifically, we show that strong geographical proportionality between population counts derived from digital trace data and from the census is not sufficient to establish representativeness, because residual bias may still vary systematically across areas and covariates.

Action: In response:

- We have revised the "Results" section to more consistently distinguish between overall coverage levels, as captured via the population coverage bias as defined in Equation (3.1) and representativeness across demographic and socioeconomic groups. Particularly, we have added the following text in "Results - The extent of population coverage bias varies across data sources": *"However, we note that broader coverage in MPD should not be interpreted as equivalent to representativeness, since a source may capture a larger share of the population overall while still systematically over or underrepresenting particular groups or areas.". We have also added the following paragraph in "Results - Population coverage bias varies widely over space": "Fourth, scatter plots indicate strong global geographic proportionality between MPD-derived and census population counts. For each source, the fitted proportional baseline $P_i^D = \hat{\alpha} P_i$ yields high R^2 with $\hat{\alpha}$ values summarising average platform coverage relative to census population. Under conventional validation approaches, high R^2 and $\hat{\alpha}$ are often interpreted as evidence of representativeness (Iradukunda et al. 2025; Renninger and Cabrera 2025). However, as shown in the next subsection, analysis of residuals around this baseline reveal considerable local variation. This indicates that ensuring that assumptions of linear spatial proportionality have been met is not sufficient to assess representativeness."*
- We have revised the "Discussion" to explicitly state that: *"We demonstrate that broader population coverage in a given source is not equivalent to better representativeness. A source may capture a larger share of the population and still systematically over or underrepresenting particular demographic, socioeconomic or geographic groups."*

R2.5: *Some figure captions and labels could be made clearer. Figure 1 is difficult to interpret due to the dual axis and unclear mapping between axes and legend. In Figure 3, "population coverage bias" appears to be on a 0-100 scale, but the wording ("size of the bias") suggests higher values are worse. If values closer to 100 indicate higher coverage, the terminology and caption should make the direction of interpretation explicit. Relatedly, page 11, line 55 states that mobile phone app data have strong potential for large-scale empirical analysis; this would read better with a short qualification that coverage/volume alone do not resolve known measurement issues linked to how these data are collected and processed.*

Action: We have taken the following actions:

- We have revised the referred Figure 1 (which is Figure 2 in the manuscript) with the dual-axis. We removed one axis and updated the relevant text: *"Figure 2 presents these comparisons across relevant data sources, showing the corresponding measure of population coverage bias on the x-axis. Higher coverage bias values (closer to 100) indicate lower population coverage relative to the census benchmark, whereas lower bias values indicate higher coverage."*
- We have updated the caption of Figure 3. We extended the caption to indicate that higher coverage bias values indicate lower population coverage.
- We have revised the statement regarding the potential of mobile phone app data for large-scale empirical analysis. It now reads: *"Overall, these results suggest that in terms of raw*

size and breadth, MDP have strong potential to support large-scale empirical analyses, although higher coverage and sample volume do not guarantee representativeness issues.".

R2.6: *The reported coverage levels for multi-source app data also need clearer support. The paper refers to high coverage values (e.g., 70% and 26% for an undisclosed provider), which appear high compared with common estimates for these datasets (often single-digit percentages or around 10%, depending on definition and time window). The paper should state exactly what these percentages represent (time window, unique active users, de-duplication, denominator) and provide evidence or careful qualification, particularly for the 26% figure.*

A: Thank you for the opportunity to clarify. The cited percentages refer to the count of unique devices in March 2021 over the total residence population at the start of the data description section. They are not the same as the percentages we report and use for our analysis. The number of unique devices we used in our analysis is smaller. The original sample reduces to a smaller subset for which repeated observations exist to identify their home location. Hence, we report population coverage on the order of 2%-8% of the census population in individual LADs. We have revised the manuscript to explain these differences.

Action:

- We revised the "Introduction" to clarify that the 70% and 23% adoption rates for Facebook and Twitter respectively are reported figures (not the same as the resident population numbers that we estimate for our analysis).
- We added text to the beginning of the section "Data - Multi-app1". It reads: *"We sourced data from a location analytics company that collects anonymised GPS data from smartphone applications. The raw dataset includes unique observed devices equivalent to approximately 26% of the UK population in March 2021. This figure is not the same as the population coverage metric we use for our analysis. The number of unique devices we used in our analysis is smaller. The original sample reduces to a smaller subset for which repeated observations exist to identify their home location."*

R2.7: *Processing choices for Twitter/X should be described. Twitter geolocation can differ between GPS geotags and platform-assigned locations, and these can conflict; the paper should state how this was handled because it affects any bias/coverage comparisons for X/Twitter.*

A: In this study, we use an anonymised, analysis-ready dataset in which user locations were inferred following the workflow described by Wang et al. 2022. In that workflow, tweet locations are derived from GPS geotags when available and complemented using place-based names collected via the Twitter/X API. We do not reprocess raw tweet locations ourselves. For Twitter/X, we rely on the published preprocessing and home-assignment procedures. We have added text to clarify this in the manuscript and explain more explicitly that the Twitter/X results should be interpreted in light of the underlying geolocation and home-detection workflow used to construct the dataset.

Action: We have added the following text in the "Data - Twitter/X" section: *"We rely on the processed dataset released by (Wang et al., 2022). We do not re-estimate tweet-level locations in this study. Wang et al.(2022) derived location information from GPS geotags when available and from place name information referring to geographic bounding boxes provided by the Twitter/X API.*

R2.8: *The interpretation of results would be stronger with more engagement with related studies, particularly in the introduction and discussion, for example:*

- <https://doi.org/10.1038/s41597-024-03490-y>
- <https://doi.org/10.1016/j.apgeog.2023.102997>
- <https://doi.org/10.1016/j.apgeog.2024.103260>
- <https://doi.org/10.1140/epjds/s13688-023-00447-w>
- <https://doi.org/10.1111/geoj.70033>
- <https://doi.org/10.3390/app15020982>
- <https://doi.org/10.1371/journal.pone.0294430>

A: We agree that engaging more deeply with this literature strengthens the positioning of our work and situates our findings within the broader field. In the revised manuscript, we have incorporated discussion of these studies where they align with our conceptual framing and empirical results. This has allowed us to …

Action: We have taken the following actions:

- We have expanded the Introduction to cite and briefly summarise these recent contributions, emphasising how they relate to: (a) the need for systematic bias assessment, and (b) the limitations of existing validation practices that focus primarily on correlations or single-source analyses.
- We have integrated more detailed engagement with these studies in the Discussion, drawing clear connections between our findings and insights from the suggested literature, particularly in relation to (REVISE!) spatial heterogeneity in bias and coverage patterns, methodological innovations for bias diagnostics and the implications of digital divide and demographic composition for representativeness.
- We point out how our results align with or extend findings from these studies — for example, in demonstrating the complexity of nonlinearity in bias drivers, and in showing that multi-app data may not automatically reduce coverage bias despite higher overall penetration.
- All suggested references have been added to the bibliography and are appropriately connected to specific claims or components of the framework (e.g., measurement, spatial clustering, explainable modelling).

R2.9: *The ethics section could also be clearer in plain terms, ideally within the paper rather than only as a brief statement. The current wording says approval was not required because no personal identifiers were processed and that the work followed University of Liverpool ethical standards. Given the use of Twitter Academic API data, estimation of home area from high-resolution traces, and processing of at least one multi-source dataset, the paper should briefly* *describe the safeguards in place (secure access, aggregation/suppression rules, limits on outputs) and whether the work was reviewed or formally exempted, noting that re-identification risk can still exist even when data are de-identified.*

A: We added text to more clearly state that all analyses were conducted under institutional data governance standards at the University of Liverpool. The datasets accessed were either: i) publicly available in processed and aggregated formats (including the datasets from the Twitter Academic API, Locomizer and Meta), or ii) under a formal data-sharing agreement that restrict access to pseudo-anonymised records and prohibit attempts at re-identification. For case i), individual-level identifiers were not accessible to the research team and in the case of Meta privacy protections (e.g., small-count suppression, thresholding, spatial smoothing, or pre-processed home assignment) were applied by the provider before data access. For case ii), outputs were generated only at aggregated geographic levels (LAD or higher) to avoid reidentification in compliance with our data sharing and non-disclosure agreements.

Actions:

- We have revised the text to explicitly state the ethical standards that we followed in the body of the manuscript. We specifically revised the beginning of the "Data" section and added the following paragraph: *"The research was conducted in compliance with University of Liverpool ethical standards. No personal identifiers were accessible to the researchers at any stage of the study. All data sources used in this analysis were stored and accessed within a secure data environment protected by three-factor authentication. All data were anonymised and when individual-record levels were available, data-sharing agreements did not permit any attempt at re-identification. All reported outputs are presented only at aggregate spatial and temporal scales. Although the analysis did not involve directly identifiable personal data, these privacy-preserving measures were implemented to minimise disclosure risk when working with mobility and social media data."*
- We have expanded the "Ethics statement". It now reads as follows: *"This research was conducted in accordance with the ethical standards of the University of Liverpool. No personal identifiers were accessible to the researchers at any stage of the study. All data used were anonymised, and Multi-app1 device-level records, together with the other data sources used in the analysis, were stored and accessed within a secure data environment protected by three-factor authentication. Data-sharing agreements did not permit any attempt at re-identification, and all reported outputs are presented only at aggregate spatial and temporal scales to minimise residual disclosure risk."*

R2.10:. *The data/code availability statement should be clearer and aligned with journal requirements. The manuscript points to GitHub, but it is unclear whether data from Locomizer and the undisclosed provider will be shared (and Table 1 suggests they may not). Given TOP standards (Data transparency Level 2: data posted to a trusted repository; exceptions identified at submission), the paper should state what will be publicly available and where. If the underlying data cannot be shared due to licensing, it should state if anything be shared* *instead (e.g., aggregated outputs or a reproducible pipeline for authorised users) and how any exception was handled at submission.*

A: We have revised the manuscript to describe the content that will become publicably accessible on our Github repository. We state that all the code and data to reproduce the reported results in the paper will be made available, including references on how to access the original sources of data. Sensitive code involving the preprocessing of our restricted-access Multi-app2 data will not be shared. All the data that will be shared will be in aggregated, anonymised format. In compliance with data sharing licenses and agreements, we will not share the original records we used for the analysis. The Twitter/X and Multi-app2 datasets used in our analysis are publicly available in analysis-ready form through the repositories cited in the manuscript. Meta data is accessible through Meta's Data and AI for Good programme. Multi-app1 is subject to a non-disclosure agreement with the provider and we can only share aggregate data outputs. Specifically we will provide a derived aggregated LAD-level dataset containing census population counts, digital-trace population counts and the computed population coverage bias used as input in our analysis.

Actions:

- We revised the "Data/code availability statement" to identify which of the data sources used in our analysis is publicly available, accessible through data provider or restricted by licensing or non-disclosure agreement. The revised text now reads as follows: *"Code and data to reproduce the reported results in the manuscript are openly available at [DEBIAS detection repository](https://github.com/de-bias/bias-detection). Sensitive code involving the preprocessing of restricted-access data is shared. Analytical workflows reported in the manuscript rely on aggregated inputs. The Twitter/X dataset used in this study is publicly available in aggregated analysis-ready form from the repository [github.com/Twitter-Internal-Migration](https://github.com/c-zhong-ucl-ac-uk/Twitter-Internal-Migration), and Multi-app2 is publicly available via [Zenodo](https://doi.org/10.5281/zenodo.13327082). Meta data were accessed through Meta's [AI for Good programme](https://ai.meta.com/ai-for-good/datasets/) and are subject to Meta's access conditions. Multi-app1 is subject to a non-disclosure agreement with the provider and cannot be shared in its raw format. To support transparency and reproducibility, we provide the processed dataset aggregated at Local Authority District level, containing census population counts, digital-trace population counts for each source, and the corresponding population coverage bias used in the analysis. It is available at [DEBIAS repository](https://github.com/de-bias/bias-detection/blob/main/outputs/manuscript-data/lad-derived-population-counts.csv).".*
- We also added text in the "Data" section indicating that we will share a publicly available derived aggregated LAD-level dataset containing census population counts, digital-trace population counts by source and the computed population coverage bias used in the analysis. The added text reads: *"A dataset aggregated at LAD-level is publicly available on [DEBIAS repository](https://github.com/de-bias/bias-detection/blob/main/outputs/manuscript-data/lad-derived-population-counts.csv). The*

dataset contains the core inputs required by our framework, including census population counts, digital-trace population counts for each source and the corresponding population coverage bias used in the analysis.".

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